

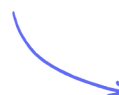
Non-Calculator Questions

Algebraic Fractions

Simplifying Algebraic Fractions / Adding & Subtracting Algebraic Fractions /
Multiplying & Dividing Algebraic Fractions / Solving Equations with Algebraic
Fractions

Medium (7 questions)	/18
Hard (8 questions)	/26
Very Hard (3 questions)	/12
Total Marks	/56

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Medium Questions

1 Simplify $\frac{p}{2q} \times \frac{4pq}{t}$.

(2 marks)

2 Write as a single fraction in its simplest form.

$$\frac{1}{x+2} - \frac{2}{3x-1}$$

(3 marks)

3 Simplify $\frac{ab - b^2}{a^2 - b^2}$.

(3 marks)

4 Write as a single fraction in its simplest form.

$$\frac{1}{x} - \frac{1}{x+1}$$

(3 marks)

5 Simplify.

$$\frac{3+x}{9-x^2}$$

(2 marks)

6 Write as a single fraction in its simplest form.

$$\frac{1}{y-1} - \frac{1}{y}$$

(3 marks)

7 Solve.

$$\frac{3}{x-2} = \frac{5}{x+1}$$

(2 marks)

Hard Questions

- 1 Write as a single fraction in its simplest form.

$$\frac{3}{x-4} - \frac{4}{x+3}$$

(3 marks)

- 2 Simplify.

$$\frac{x^2 - 5x}{2x^2 - 50}$$

(4 marks)

- 3 Write as a single fraction in its simplest form.

$$2 - \frac{2x - 1}{x + 1}$$

(3 marks)

- 4 Write as a single fraction in its simplest form.

$$\frac{x+3}{x-3} - \frac{x-2}{x+2}$$

(4 marks)

5 Write as a single fraction in its simplest form.

$$\frac{2x}{x+3} + \frac{x+3}{x-5}$$

(3 marks)

6 Simplify $\frac{x^3 + 5x^2}{x^2 - 25}$, giving your answer as a single fraction.

(3 marks)

7 Write as a single fraction in its simplest form.

$$\frac{5}{x-2} - \frac{x-5}{2}$$

(3 marks)

8

$$\frac{3}{m+4} - \frac{4}{m} = 6$$

Show that this equation can be written as $6m^2 + 25m + 16 = 0$.

(3 marks)

Very Hard Questions

1 Simplify.

$$\frac{ux - 2u - x + 2}{u^2 - 1}$$

(4 marks)

2 $f(x) = 3x + 2$ and $g(x) = x^2 + 1$.

Find $\frac{g(x)}{f(x)} + x$.

Give your answer as a single fraction, in terms of x , in its simplest form.

(3 marks)

3 (a) Simplify fully $\frac{3 - x}{3x^2 - 5x - 12}$

(2 marks)

(b) Write $\frac{x}{x-1} - \frac{x}{x+1}$ as a single fraction in its simplest form.

(3 marks)